

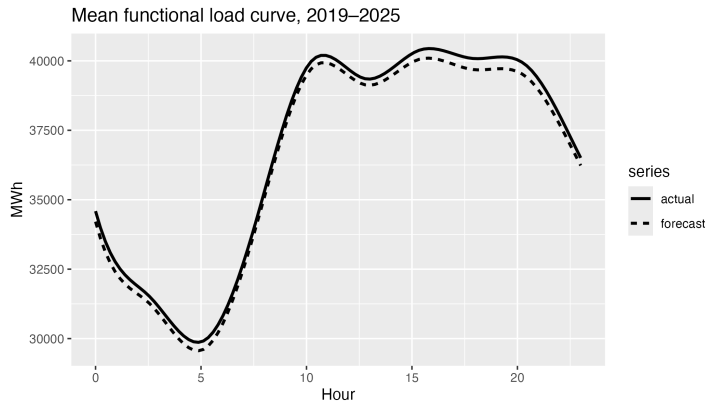
Functional Forecasting of Turkish Electricity Load Curves

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Thesis Proposal

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The Forecasting Object: A Daily Load Curve



$$Y_d(h), \quad h = 0, \dots, 23$$

- A single **curve**, not 24 scalars
- Dual-peak shape is stable across all days
- Solid = actual; dashed = official forecast
- **Goal:** forecast tomorrow's full 24-hour profile

Why Accuracy Matters

Key users (post-2013 liberalisation):

- **TEİAŞ** (system operator)
unit commitment, grid scheduling
- **Generators**
day-ahead market bids on EPIAS
- **Large consumers**
real-time imbalance penalties

Systematic underprediction — official TEİAŞ forecast

Year	Official bias (MWh/hour)	Our bias (MWh/hour)
2023	+344	+176
2024	+585	+34
2025	+472	+98
Average	+467	+103

Positive bias = forecast < actual (underprediction).

Research Questions

- ① **Representation:** Can B-spline smoothing and FPCA represent Turkey's daily electricity load curves with negligible information loss?
How many functional principal components suffice?
- ② **Forecasting accuracy:** Does a functional VAR model augmented with holiday and derivative features statistically outperform the official TEİAŞ day-ahead forecast?
Tested by Diebold-Mariano over 1,096 hold-out days, 2023–2025.
- ③ **Error structure:** Can a data-driven residual correction further reduce errors at the morning ramp hours where the base model underperforms?
Does serially correlated error contain exploitable signal?

Literature Positioning [to be expanded]

Functional Data Analysis

- Ramsay & Silverman (2005): FDA foundations
- Kokoszka & Reimherr (2017): modern inference
- Hyndman & Ullah (2007): functional forecasting (demographics)

FDA applied to load curves

- *[to be identified in literature review]*

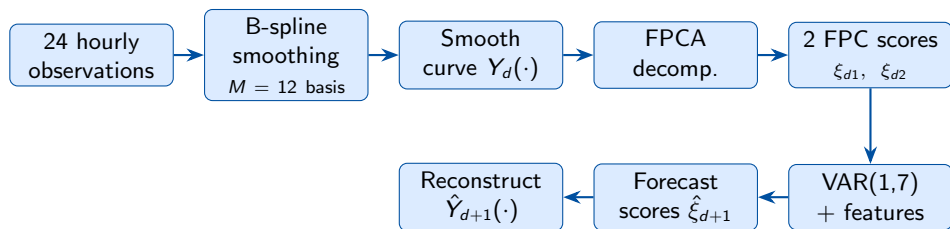
Short-term electricity forecasting

- Classical: ARIMA, exponential smoothing
- ML: gradient boosting, neural networks
- Xie & Hong (2016): residual simulation / correction

Gap this thesis fills

- No FDA study on Turkey's post-liberalisation electricity market
- Holiday + derivative features tailored to Turkish market structure
- Formal DM testing against operational benchmark

From 24 Numbers to a Curve: The FDA Pipeline



B-splines

preserves shape; enables differentiation

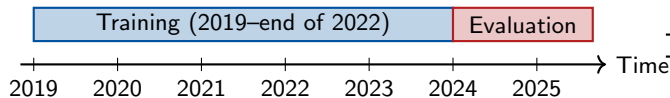
FPCA

2 scores carry 97.3% of curve variation

VAR(1,7)

weekly + daily score dynamics

Data



Rolling-origin design: model re-estimated before each forecast day using only past data. See evaluation slide.

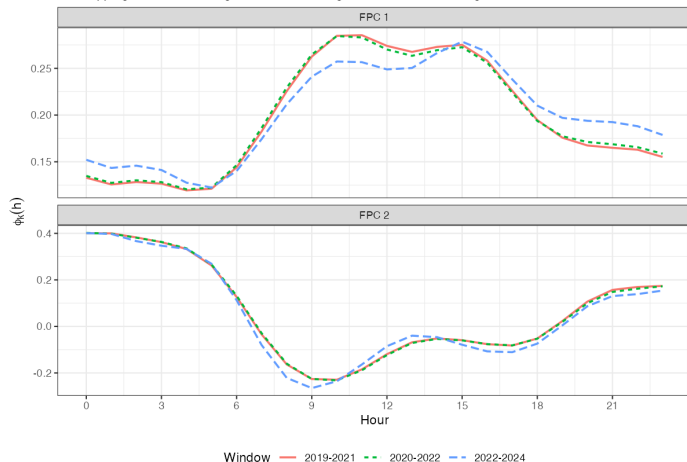
Series	Source
Hourly load (MWh)	EPIAS
Official forecast	TEİAŞ / EPIAS
Holiday calendar	Official gazette

Item	Value
Frequency	Hourly
Coverage	2019–2025
Training curves	≈1,460 days
Evaluation curves	1,096 days
Hourly errors eval	26,304

FPCA: Two Numbers Capture the Curve

FPCA eigenfunctions across sub-periods

Overlapping lines = stable eigenfunctions; divergence = structural change



FPC	Variance	Cumulative
1	93.4%	93.4%
2	3.9%	97.3%
3	1.1%	98.4%

FPC 1 — Level

Nearly flat $\Rightarrow$ shifts the whole curve up or down

FPC 2 — Shape

Positive overnight & late evening, negative daytime $\Rightarrow$ overnight-vs-daytime contrast

Stability: three sub-period fits nearly coincide $\Rightarrow$ static FPCA justified

Robustness: $K = 4$ adds $< 1\%$ further variance; results unchanged

Score Forecasting Model

Core score dynamics:

$$\xi_d = a + A_1 \xi_{d-1} + A_7 \xi_{d-7} + B z_d + u_d$$

- **Lag 1:** yesterday's level and shape
- **Lag 7:** same-day-last-week seasonality
- z_d : feature vector (all predetermined)

Reconstruction:

$$\hat{Y}_d(h) = \hat{\mu}(h) + \hat{\phi}_1(h) \hat{\xi}_{d1} + \hat{\phi}_2(h) \hat{\xi}_{d2}$$

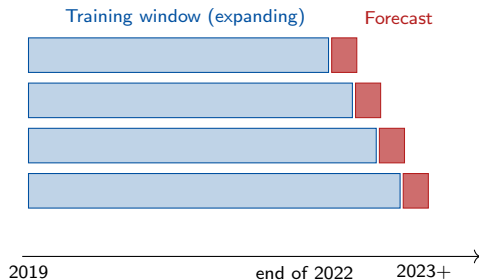
Estimated by OLS, equation-by-equation.

Feature blocks in z_d :

Block	Content
Calendar	Weekend, annual sine/cosine
Holiday	Public holidays, Eid, bridge days
Load shape	Yesterday's mean, peak, range
Rolling avg.	7-day, 28-day averages
Derivative	Morning ramp, evening ramp, slope volatility

Holiday and derivative blocks are the critical additions — without them the model does not beat the official.

Evaluation: Rolling-Origin Design



Each forecast uses only data dated before that day.
FPCA and VAR both re-estimated at every step.

Metrics:

$$\text{MAE} = \frac{1}{N} \sum_{d,h} |e_{dh}|$$

$$\text{MAPE} = \frac{1}{N} \sum_{d,h} \frac{|e_{dh}|}{Y_d(h)}$$

Significance test:

Diebold-Mariano (1995)

H_1 : challenger MAE < official MAE

Pooled (26,304 obs.) + hour-by-hour

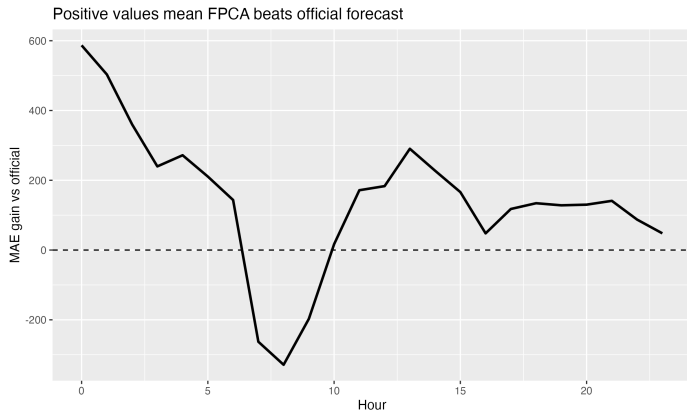
A lower MAE number alone is not sufficient evidence.

Main Results: DM Confirms the Gains

Model	MAE	MAPE	vs. official	DM
Seasonal naïve Y_{d-7}	1912	5.13%	−57.6%	—
Official (TEİAŞ)	1214	3.15%	ref.	—
FPCA VAR(1,7)	1443	3.76%	−18.9%	n.s.
+ calendar / ramp	1238	3.26%	−2.1%	n.s.
+ holiday / load-shape	1001	2.63%	+17.5%	***
+ derivative features	980	2.56%	+ 19.2%	***
+ residual correction	653	1.70%	+ 46.2%	***

DM test vs. official: *** $p < 0.001$; n.s. = not significant. MAE in MWh/hour; pooled over 26,304 obs.

Where the Model Wins and Where It Loses



Positive = model beats official (MAE, MWh)

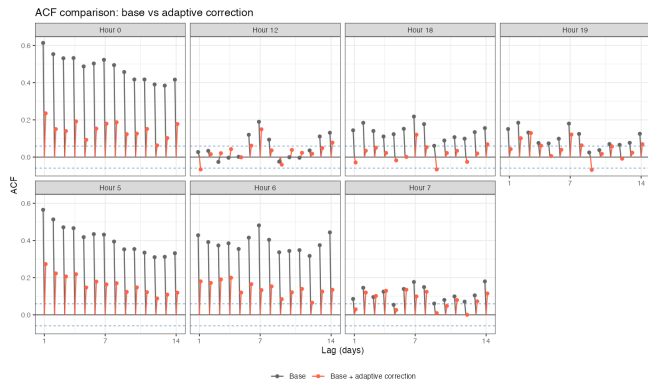
Wins: hours 0–5 and 10–23
(overnight trough and daytime)

Loses: hours 6–9
(morning ramp-up)

The source of the official forecast's ramp-hour advantage is not known. Our model uses historical load patterns only.

DM: significant at 18 of 24 hours (5% level).

Residual Correction: Exploiting Persistent Errors



Motivation: errors are serially correlated
Ljung-Box rejects at all 24 hours

$$\hat{Y}_d^{\text{res}}(h) = \hat{Y}_d^{\text{base}}(h) + \lambda_d \hat{r}_d(h)$$

$\hat{r}_d(h)$: correction curve from recent residuals

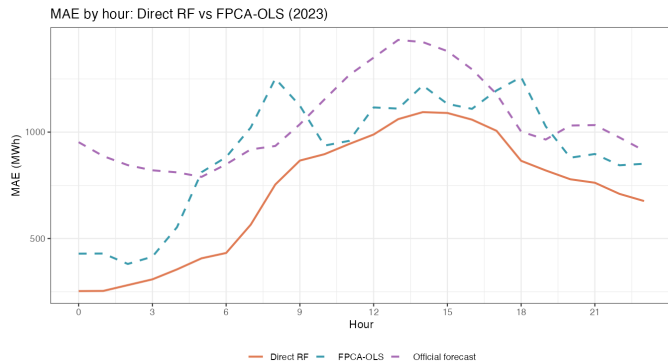
λ_d : rolling CV on pre- d data only

Result:

- +46% MAE vs. official
- Wins at **all 24 hours** (incl. ramp hours)

Gray = base model ACF; Orange = after correction

Random Forest Benchmark: Strong but Different



2023 benchmark sample

8,760 hourly forecasts

Model	MAE	vs. off.	DM
Official	1052	ref.	—
FPCA-OLS	910	+13.5%	***
Direct RF	718	+31.8%	***

RF vs. FPCA-OLS:

Direct RF has lower MAE at **24/24 hours**;
pairwise DM $p \approx 1.2 \times 10^{-85}$.

Interpretation:

- Nonlinear hour-specific models are a serious benchmark.
- FDA remains the interpretable functional baseline, evaluated over 2023–2025.

*** $p < 0.001$ vs. official.

Summary of Contributions

Methodological

- Functional representation of Turkey's daily load curves
B-splines + FPCA; $K = 2$ retains 97.3% of variation
- Holiday-aware and derivative-augmented feature design for functional VAR
- Residual correction via rolling CV — zero lookahead

Empirical

- Base model: **+19.2%** MAE vs. official TEİAŞ
DM $p \approx 10^{-163}$; 1,096 days, 2023–2025
- Residual correction: **+46.2%**; wins at all 24 hours
- Average bias reduced from 467 to 103 MWh/hour
- Random Forest benchmarked as nonlinear reference

Core claim: Functional representation combined with domain-aware features yields statistically significant, operationally relevant improvement over the official benchmark.

Thesis Road Map

Completed

- ✓ Functional representation (B-splines, FPCA)
- ✓ Rolling-origin evaluation framework
- ✓ VAR(1,7) + feature blocks
- ✓ Residual correction step
- ✓ Diebold-Mariano testing (pooled + hourly)
- ✓ Random forest benchmark
- ✓ Eigenfunction stability analysis, $K = 4$ check

Planned

- ☐ Systematic literature review
- ☐ Ramp-hour error diagnostics
- ☐ Probabilistic forecasting (prediction intervals)
- ☐ Thesis write-up

References

- Kokoszka, P. & Reimherr, M. (2017). *Introduction to Functional Data Analysis*. CRC Press.
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- Hyndman, R.J. & Ullah, M.S. (2007). Robust forecasting of mortality and fertility rates: a robust functional data approach. *Computational Statistics & Data Analysis*, 51(10), 4942–4956.
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